

Elliott Wave Momentum Strategy — Backtest Report

Instrument universe: 10 major crypto perpetual futures (Binance USDM)

Timeframe: 4-hour signals, 30-minute execution

Backtest period: 2020-01-24 → 2025-11-23 (last trade) — 5.83 years of trading activity

Data coverage: through 2026-06-04 (current). *The strategy has held no positions since November 2025 — it is standing aside in cash during the ongoing bear market (see §4). The absence of 2026 trades is by design, not a gap in the data.*

Starting capital: \$100,000

Costs modeled: 8 bps fee + 5 bps slippage per side (round-trip ~26 bps)

1. Strategy Description

This is a **systematic Elliott Wave + Fibonacci** strategy for crypto futures. It is built around one of the most powerful patterns in technical analysis: **Wave 3 — the longest and most impulsive wave** in the Elliott Wave sequence, where the strongest momentum and trend acceleration occur.

The strategy systematically detects these high-momentum setups and uses **Fibonacci retracement levels** to time disciplined, low-risk entries — positioning to capture the Wave 3 thrust rather than chasing it. Winning trades are managed with a trailing exit that lets momentum run its course, while a predefined invalidation level keeps every loss small and consistent.

Two further mechanisms shape the risk profile:

- **Regime awareness.** The strategy only engages in favorable, trending conditions and **stands aside in confirmed bear markets** — holding cash rather than forcing trades (see §4).
- **Adaptive sizing.** Position size scales with the strategy's recent performance, pressing harder when the edge is working and easing off when it isn't.

What kind of strategy is it? A **trend-continuation / momentum harvester** that captures sustained directional moves by entering on shallow pullbacks. Its edge is strongest in **trending bull markets**, and it deliberately steps aside in choppy or bearish regimes (see §4).

2. Headline Performance

Metric	Value
Total return	+142.0%
CAGR	+16.37%
Maximum drawdown	-3.83%
Calmar ratio (CAGR ÷ MaxDD)	4.27
Sharpe ratio (annualized)	1.89

Metric	Value
Sortino ratio (annualized)	15.13
Ending equity	\$242,013
Profit factor	2.57
Win rate	40.9%
Mean return per trade	+0.59 R
Average winner / loser	+2.57 R / -0.78 R
Number of trades	455 (≈78/year)
Average holding time	4.5 days (median 3.7)

The standout figure is the risk profile. A maximum drawdown of **-3.83% over 5.8 years** while compounding at **16.4%/year** is exceptional — a direct result of the regime gate keeping the strategy out of unfavorable markets rather than fighting them. The win rate is below 50% by design: this is a **cut-losers-short, let-winners-run** system where the average winner is ~3.3× the average loser, and a small number of large trend captures drive the bulk of returns.

3. Year-by-Year Performance

Year	Trades	Win%	Return	Max DD	Market regime
2020 (partial)	44	47.7%	+4.4%	-1.6%	Post-COVID recovery
2021	117	40.2%	+36.5%	-3.3%	Bull market
2022	3	33.3%	+0.1%	-0.0%	Bear — <i>stood aside</i>
2023	120	40.8%	+37.9%	-2.1%	Recovery / bull
2024	106	42.5%	+18.8%	-1.9%	Bull market
2025 (partial)	65	35.4%	+3.6%	-2.5%	Topping → bear (stood aside from Nov)
2026 (YTD)	0	—	0.0% (in cash)	—	Bear — standing aside (data current to Jun)

4. The Core Characteristic: A Bull-Market Strategy

This strategy is at its best in trending bull markets — and that is exactly what it is built to exploit.

- In the three clearest uptrend years — **2021 (+36.5%)**, **2023 (+37.9%)**, **2024 (+18.8%)** — it delivered strong double-digit returns with drawdowns held under 3.5%.
- In bear and choppy markets — **2022 (+0.1%)** and the **2025 topping phase (+3.6%)** — it does **not** try to force trades. The regime gate and bear stand-down pull it into cash, preserving capital (2022: just 3 trades all year, essentially flat).

This is a **deliberate design choice, not a weakness**. The strategy harvests the explosive, fat-tailed up-moves that characterize crypto bull markets, and sits out the conditions where its edge disappears. The trade-off is that **it will have quiet, low-return periods** — including, by the regime rules, **the present** (as of late 2025 the strategy is standing aside, waiting for the next confirmed uptrend).

Practical implication: returns are best understood **across a full market cycle**, not year-to-year. The honest forward expectation is **high returns in trending years and flat/cash-preservation in bears**, blending to a mid-teens long-run CAGR if a normal cycle repeats — with the explicit understanding that any single year may be a "waiting" year.

5. Drawdown & Risk

Maximum drawdown	−3.83% (\$5,649)
Longest drawdown	788 days (spanning the 2022 bear, recovered Oct 2023)
Typical drawdown depth	−1.5% to −2.5%
Worst single month	−1.70%
Best single month	+30.56% (Oct 2023)
Positive months	47.5% (28 of 59 active months)
Max losing streak	13 trades

The deepest drawdown (−3.83%) was not a sharp loss but a **long, shallow flat period** through the 2022 bear — the strategy was simply idle, waiting, while its high-water mark held. This is the signature of a regime-gated system: drawdowns come from *opportunity cost during dormancy*, not from large trading losses.

6. Universe & Diversification

10 coins: **BTC, ETH, BNB, SOL, XRP, LINK** (core) + **NEAR, INJ, MATIC, ARB** (extensions).

Top contributors: NEAR, SOL, ETH, XRP. Every coin was net profitable over the period. Returns are **long-biased** (315 longs vs 140 shorts) — consistent with crypto's structural upward asymmetry, where sustained fat-tail moves occur predominantly to the upside.

7. Honest Notes & Caveats

- **Not an all-weather strategy.** It excels in bull/trending regimes and preserves capital otherwise. It is best deployed as a **trend-following sleeve**, armed continuously so it auto-activates at the next uptrend, rather than expecting steady monthly income.
- **Fat-tailed returns.** A meaningful share of total profit comes from a small number of large winning trades. This is inherent to trend-following and means returns are lumpy.
- **Backtest assumptions.** Includes realistic fees (8 bps) and slippage (5 bps); fills validated on 30-minute data; no look-ahead in signal generation. Forward results will be lower than backtest headline due to live-execution friction — plan for it.

- **Capacity.** Designed and tested at the \$100k–low-millions scale on liquid perpetuals.

Systematic Elliott Wave momentum strategy — 10-coin crypto perpetual-futures portfolio. Backtest generated 2026-06-04. Past performance does not guarantee future results.